

Network Flow based IoT Botnet Attack Detection using Deep Learning

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Introduction

- Governments around the globe are promoting smart city applications to enhance the quality of daily-life activities in urban areas.
- Smart cities include internet-enabled devices that are used by applications like health care, power grid, water treatment, traffic control, etc.
- The number of Internet-of-things (IoT) based botnet attacks are increasing due to the growing of Internet-enabled devices [1] [2].

Introduction (cont'd)

- To provide advanced cybersecurity solutions to IoT devices and smart city applications, this paper proposes a deep learning (DL) based botnet detection system and also, studies the effectiveness of many machine learning (ML) classifiers and a deep learning (DL) model for botnet detection using network traffic flows.
- The botnet detection system collects the network traffic flows, converts them into connection records and uses a DL model to detect attacks emanating from the compromised IoT devices.

Objectives

- To design and train a DNN model for botnet attack detection based on network flow data.
- To train many machine learning (ML) models and comparatively analyse the performance of ML models and the deep neural network (DNN) model for botnet detection with various datasets.
- To visualize the datasets to understand its characteristics using t-distributed stochastic neighbor embedding (t-SNE) and histogram plots.

Dataset Description

- DS1 : N-BaloT [3] - 3 configurations.

Class	#Samples
Normal	172,641
Attack	240,000

DS1-V1

Botnet	#Samples
BASHLITE	135,000
Mirai	105,000

DS1-V2

Dataset Description (cont'd)

Botnet	Attack	#Samples
BASHLITE	combo	30,000
	junk	24,000
	scan	27,000
	tcp	26,999
	udp	27,000
Mirai	ack	21,000
	scan	21,000
	syn	21,000
	udp	21,000
	udpplain	21,000

DS1-V3

Dataset Description (cont'd)

- DS2 : BoT-IoT [4] - 2 configurations.

Class	Train	Test
Normal	370	107
Attack	2,934,447	733,598
Total	2,934,817	733,705

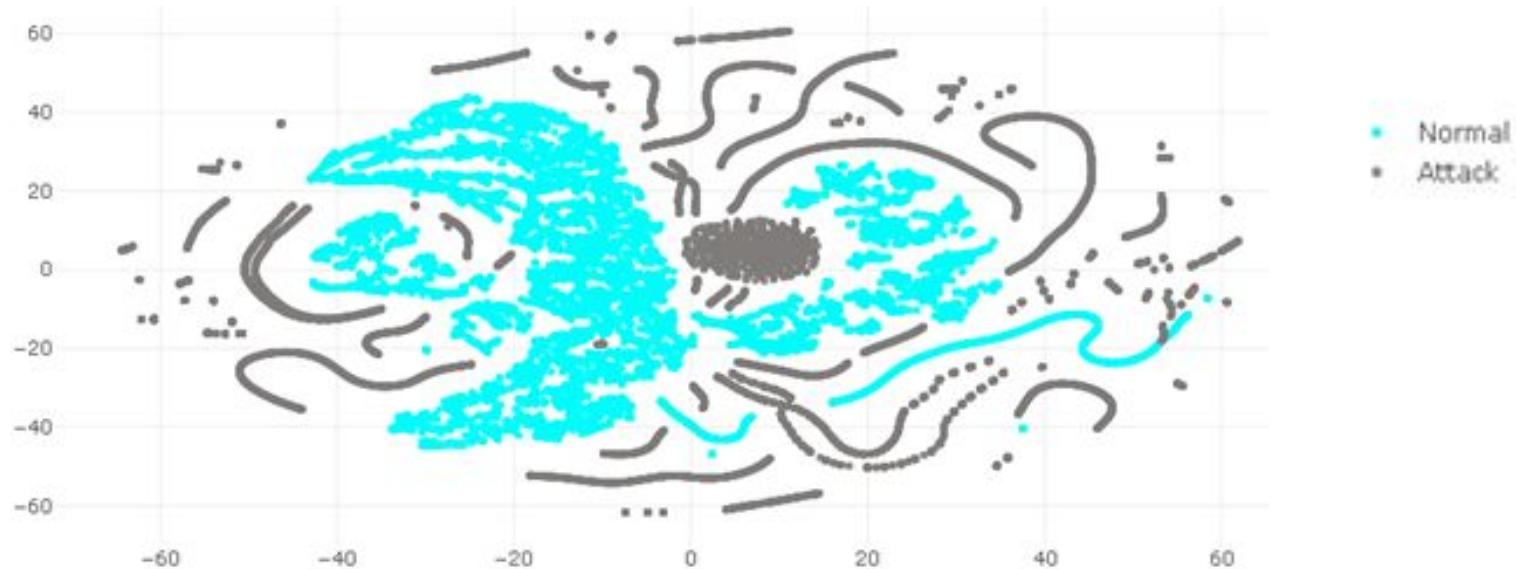
DS2-V1

Class	Train	Test
DDoS	1,541,315	385,309
DoS	1,320,148	330,112
Reconnaissance	72,919	18,163
Theft	65	14
Total	2,934,447	733,598

DS2-V2

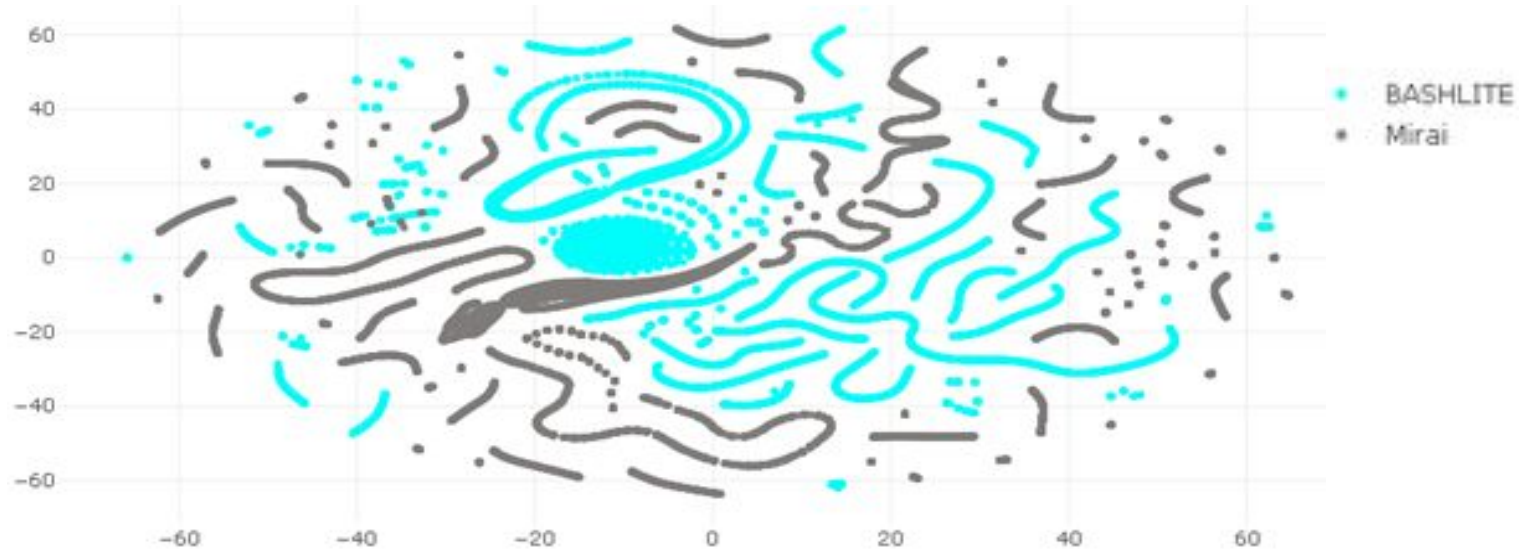
Dataset visualization

- All three configuration of DS1 dataset is visualized using t-SNE.



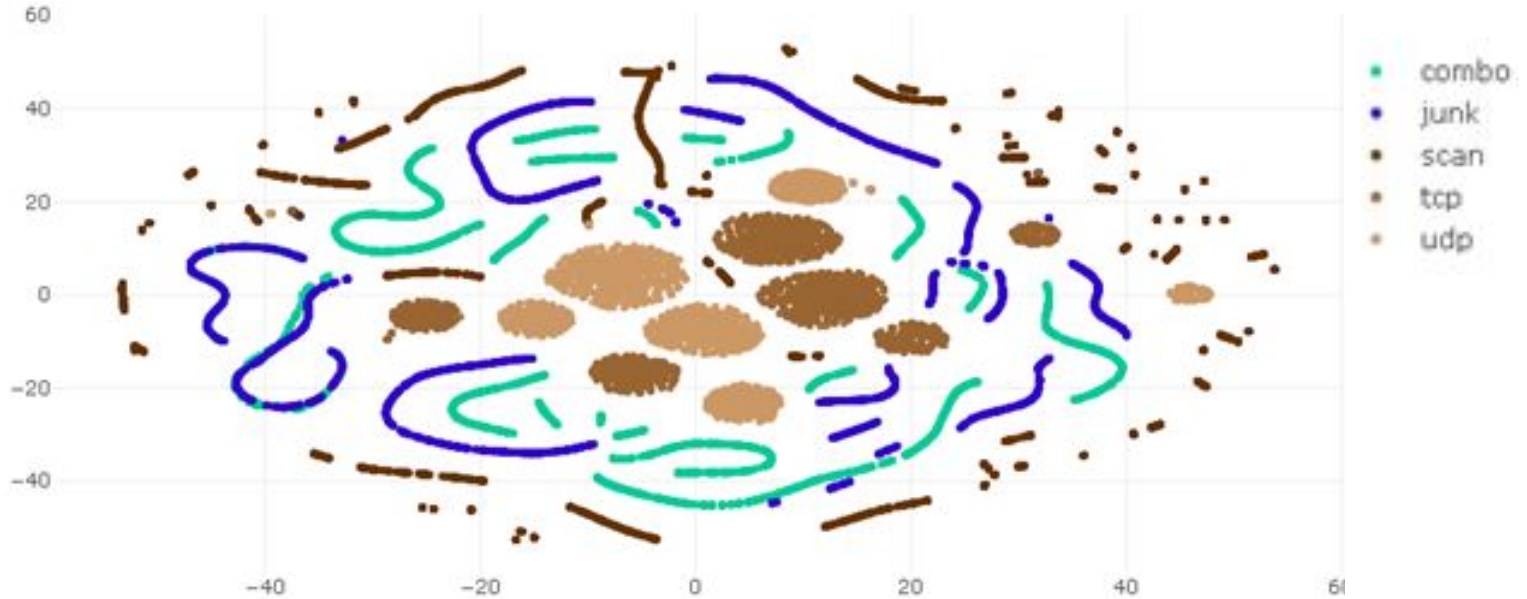
DS1-V1 visualization (normal and attack data)

Dataset visualization (cont'd)



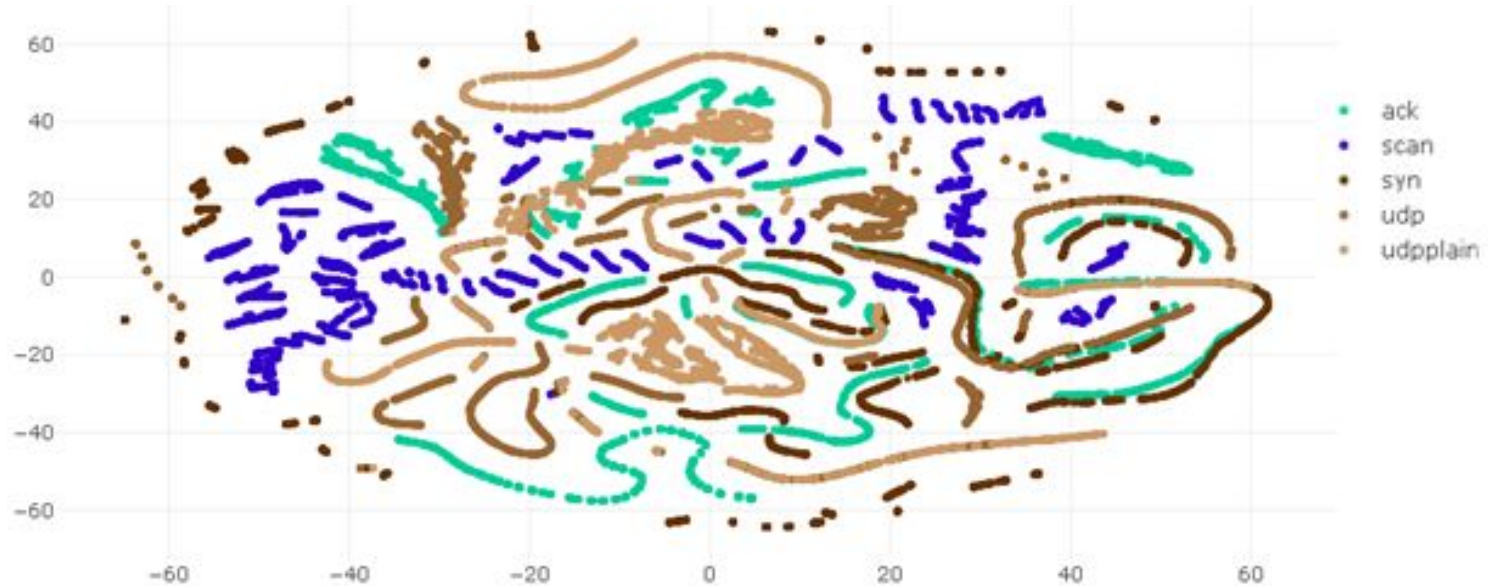
DS1-V2 visualization (Mirai and BASHLITE attack data)

Dataset visualization (cont'd)



DS1-V3 BASHLITE attack categories visualization (combo, junk, scan, tcp, udp)

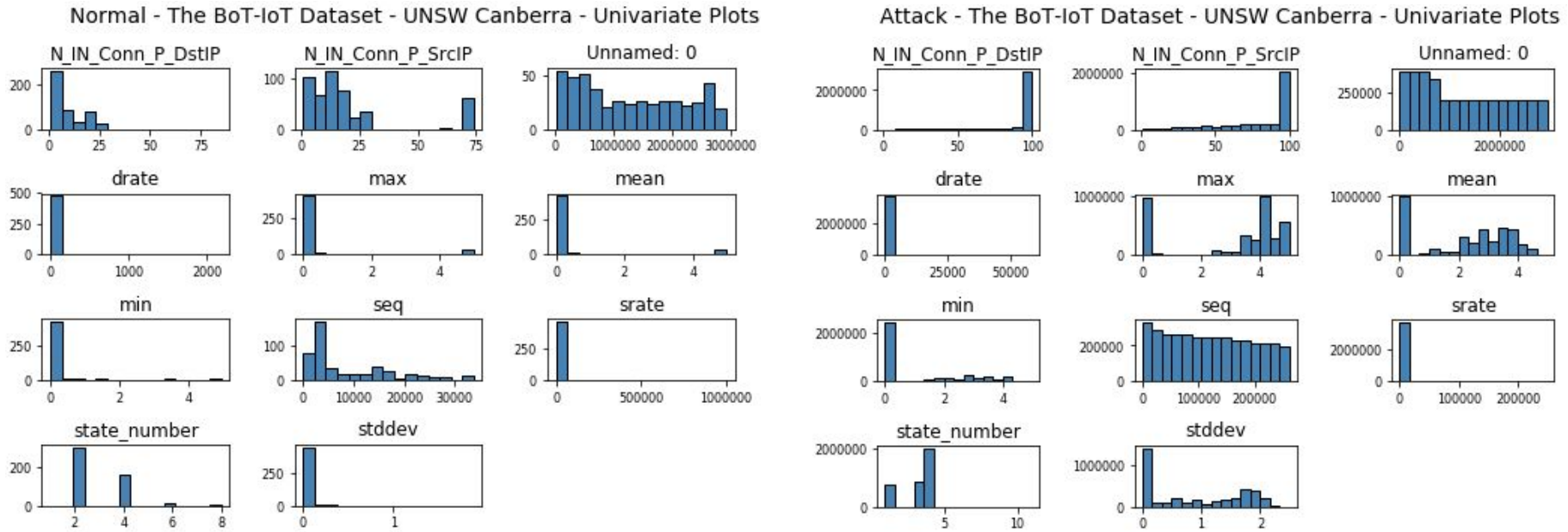
Dataset visualization (cont'd)



DS1-V3 Mirai attack categories visualization (ack, scan, syn, udp, udpplain)

Dataset visualization (cont'd)

- Histogram plots are used to represent the data distribution of DS2-V1.



DS2-V1 visualization (Normal and attack)

Methodology

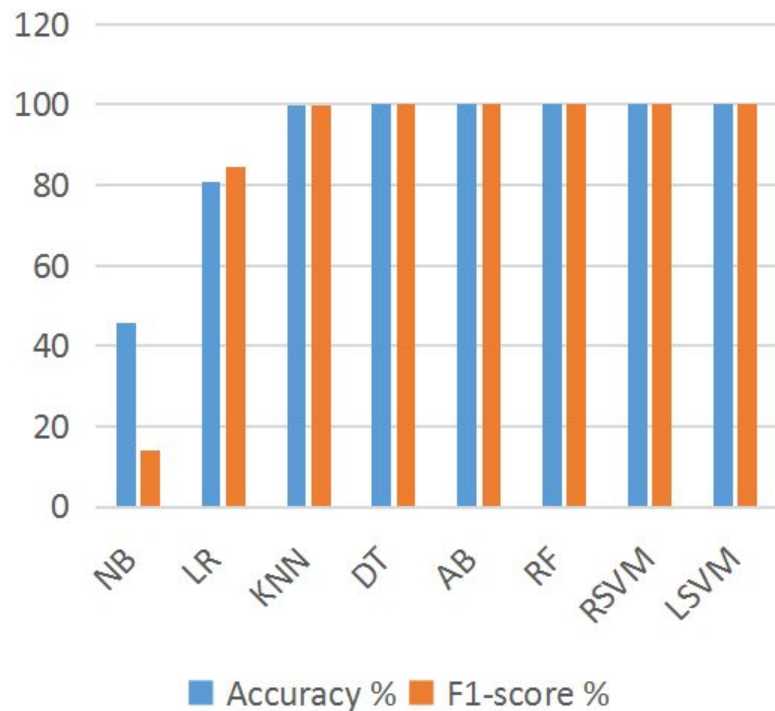
- Many DNN with the different number of hidden layers and neurons are trained and grid search is used to select optimal hyperparameters.
- The proposed DNN model has 4 hidden layers with 115, 256, 128, 64 neurons at each layer. Each hidden layer is followed by the ReLU activation function and the dropout regularization method is also employed.
- The batch size and learning rate are set to 64 and 0.001 respectively. The adam optimizer is used.

Methodology (cont'd)

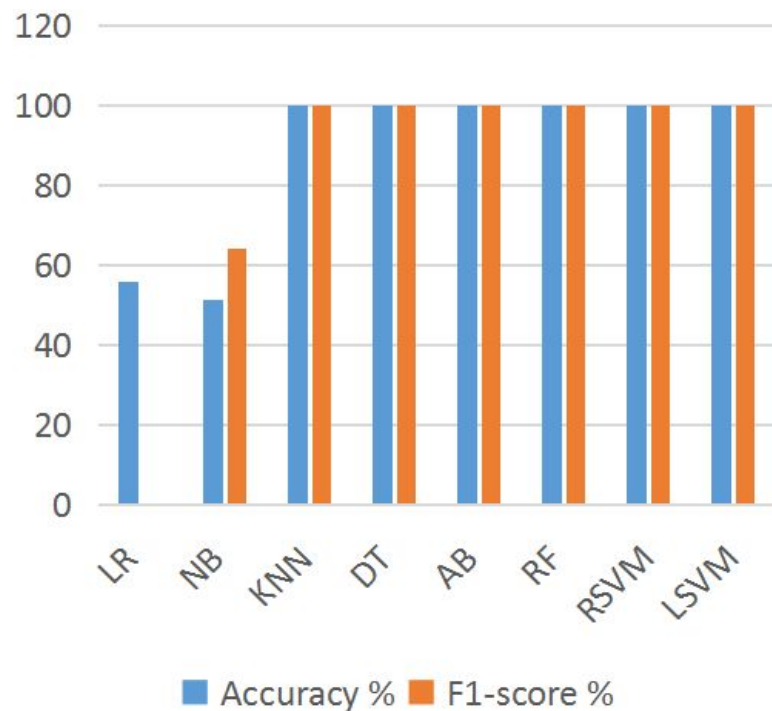
- Many ML models such as logistic regression (LR), naive bayes (NB), k-nearest neighbors (KNN), decision tree (DT), adaboost (AB), random forest (RF), Linear support vector machine (LSVM) and radial basis function SVM (RSVM) are trained.
- The default parameters set by scikit-learn¹ python library are used.

¹ <https://scikit-learn.org/stable/>

Experimental Results

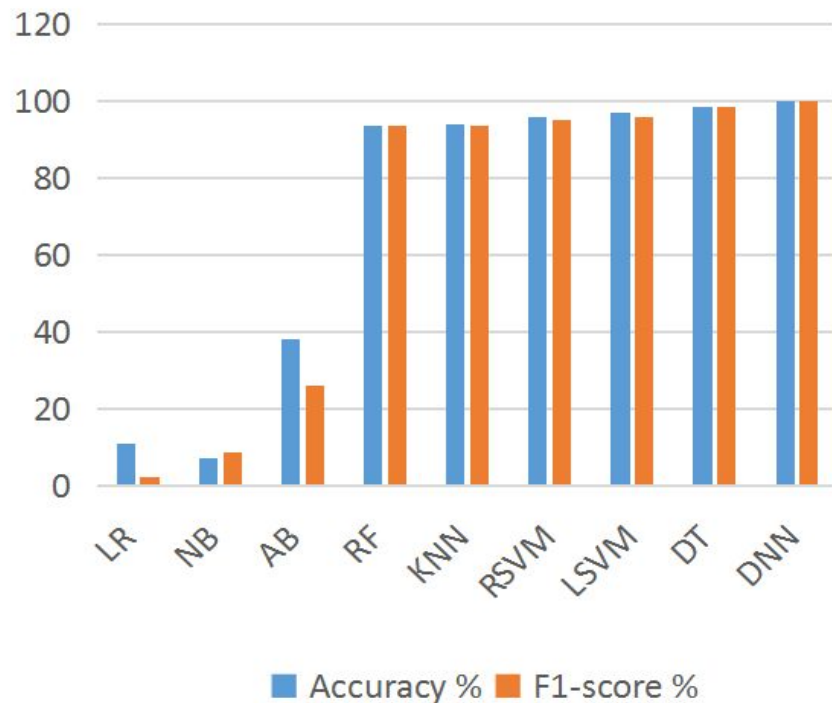


Test result - DS1-V1 (Normal and attack)



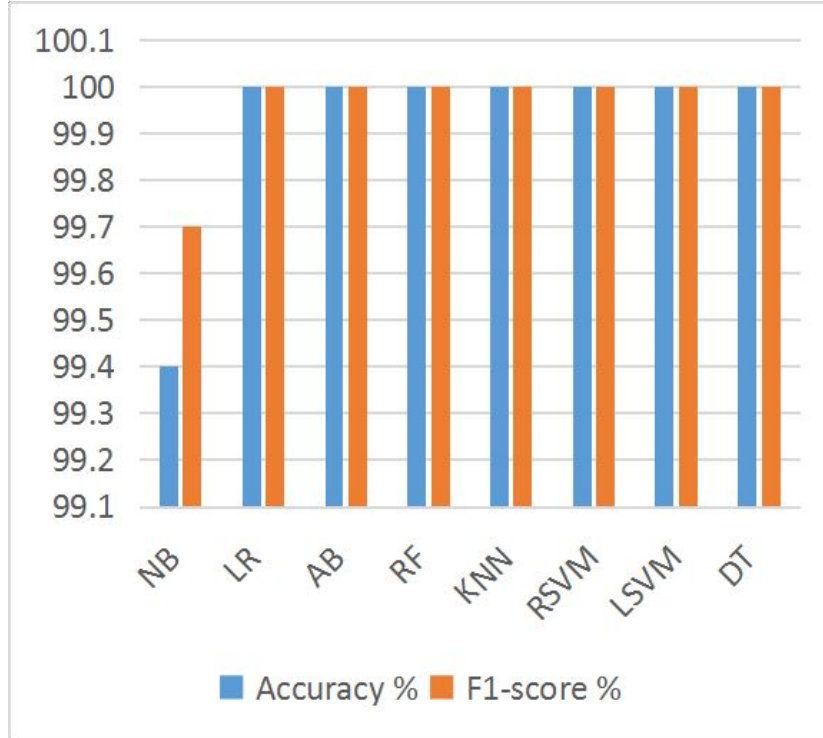
Test result - DS1-V2 (Mirai and BASHLITE)

Experimental Results (cont'd)

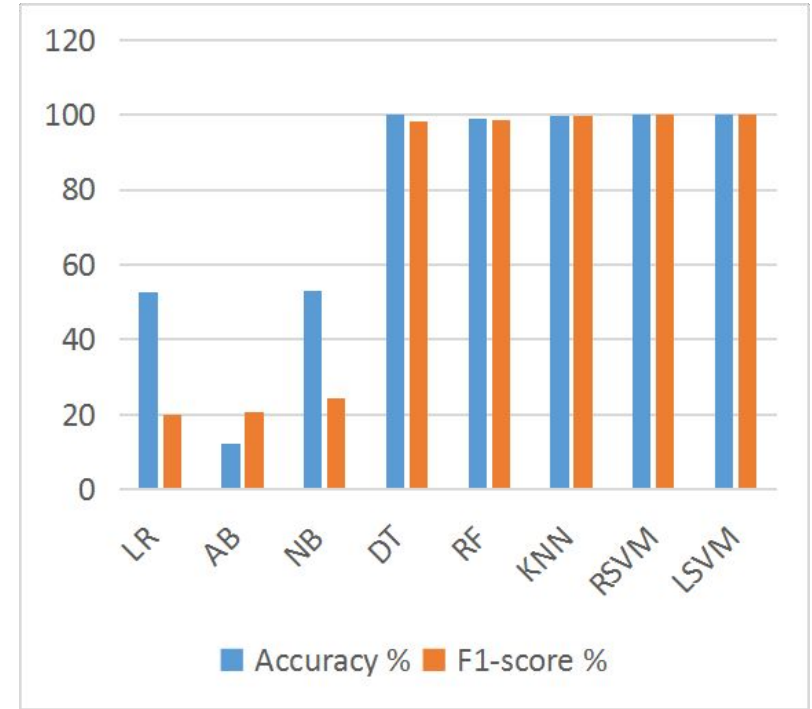


Test result - DS1-V3 (Mirai and BASHLITE categories)

Experimental Results (cont'd)



Test result - DS2-V1 (Normal and attack)



Test result - DS2-V2 (Attack Categories)

Discussion

- In case of DS1-V1 and DS1-V2, the t-SNE visualization shows that the data appears in separate clusters and looks separable. This may be the reason why most of the ML classifiers achieve perfect results.
- The DS1-V3 visualization shows some overlapping but still DNN is able to achieve 100% accurate results.
- Considering all three experiments using DS1 dataset, the DT classifier achieves the best overall performance in terms of accuracy, f1-score and also training and testing time.

Discussion (cont'd)

- The DS2-V1 histogram visualization shows that data distribution of normal and attack data are distinguishable which may be the reason why most ML classifiers achieved perfect results.
- Similar to experiments on DS1 dataset, DT classifier performed better than the rest in both the experiments on DS2 dataset. It is also computationally inexpensive.

Conclusion

- From the experiments conducted using both data sets, it can be observed that the DT is the optimal algorithm and it is generalizable across different types of data sets. It is also noticed that DNN performed better than the DT classifier in the case of experiment conducted using DS1-V3.
- This work uses features that are extracted from the network traffic flows. Instead, the entire payload data can be analyzed to distinguish between the legitimate and botnet attacks activity. Additionally, the network flow can be represented in the form of images and application CNN can be utilized for analysis and multi-modal learning can also be employed to enhance the performance of the existing works.

Thank you.

References

- [1] Akshaya, A. (2019). Massive Botnet Attack Used More Than 400,000 IoT Devices [Online]. Available at: <https://www.bankinfosecurity.com/massive-botnet-attack-used-more-than-400000-iot-devices-a-12841> (Accessed: 10 may 2020)
- [2] Zak, D. (2019). Cyberattacks On IOT Devices Surge 300% In 2019 [Online]. Available at: <https://www.forbes.com/sites/zakdoffman/2019/09/14/dangerous-cyberattacks-on-iot-devices-up-300-in-2019-now-rampant-report-claims/#4337ae2a5892> (Accessed: 10 may 2020)
- [3] Meidan, Y., Bohadana, M., Mathov, Y., Mirsky, Y., Shabtai, A., Breitenbacher, D., & Elovici, Y. (2018). N-baiot—network-based detection of iot botnet attacks using deep autoencoders. *IEEE Pervasive Computing*, 17(3), 12-22.
- [4] Koroniotis, N., Moustafa, N., Sitnikova, E., & Turnbull, B. (2019). Towards the development of realistic botnet dataset in the internet of things for network forensic analytics: Bot-iot dataset. *Future Generation Computer Systems*, 100, 779-796.

Appendix

t-SNE Details

Data set	Iteration	Error	Time in Seconds
Using no_dims = 2, perplexity = 30.000000, and theta = 0.500000			
<i>DS1 – V1</i>	1000	19.28236435	431.81
<i>DS1 – V2</i>	1000	18.85542265	431.93
<i>DS1 – V3</i> (BASHLITE)	1000	17.9763077	893.50
<i>DS1 – V3</i> (Mirai)	1000	19.53498875	636.63

Appendix (cont'd)

Experiment on DS1-V1

Model	Acc	Pre	Rec	F1	Time (seconds)	
					Training	Testing
LR	80.9	80.4	88.9	84.4	3.222	0.016
NB	45.6	88.4	7.6	14.0	0.360	0.160
KNN	99.8	99.8	99.8	99.8	2.290	46.870
DT	100	100	100	100	8.110	0.0240
AB	100	100	100	100	176.240	1.670
RF	100	100	100	100	59.320	0.740
RSVM	100	100	100	100	518,400.540	251.040
LSVM	100	100	100	100	451,215.210	185.020

Appendix (cont'd)

Experiment on DS1-V2

Model	Acc	Pre	Rec	F1	Time (seconds)	
					Training	Testing
LR	56.0	0	0	0	2.794	0.030
NB	51.3	47.5	99.7	64.3	0.269	0.104
KNN	100	100	99.9	99.9	14.614	37.181
DT	100	100	100	100	0.749	0.016
AB	100	100	100	100	0.809	0.027
RF	100	100	100	100	14.933	0.332
RSVM	100	100	100	100	501,541.460	221.800
LSVM	100	100	100	100	478,541.410	169.700

Appendix (cont'd)

Experiment on DS1-V3

Model	Acc	Pre	Rec	F1	Time (seconds)	
					Training	Testing
LR	11.1	10.3	11.1	2.3	199.484	0.096
NB	7.2	10.7	7.2	8.6	0.426	0.681
KNN	93.9	95.2	93.9	93.7	11.620	37.498
DT	98.5	98.6	98.5	98.5	58.227	1.236
AB	38.3	28.3	38.3	26.2	121.012	3.936
RF	93.6	94.9	93.6	93.4	7.695	0.026
RSVM	95.8	94.1	94.8	95.1	645,847.280	284.100
LSVM	96.8	95.7	95.1	95.9	517,114.250	210.700
DNN	100	100	100	100	412, 111.150	1.310

Appendix (cont'd)

Experiment on DS2-V1

Cl	Acc	Pre	Rec	F1	Time (seconds)	
					Training	Testing
LR	100	100	100	100	7.263	0.011
NB	99.4	100	99.4	99.7	0.413	0.090
KNN	100	100	100	100	7925.903	13.838
DT	100	100	100	100	42.048	0.052
AB	100	100	100	100	283.057	6.739
RF	100	100	100	100	884.666	5.681
RSVM	100	100	100	100	1,672,675.840	14.790
LSVM	100	100	100	100	905,893.870	5.553

Appendix (cont'd)

Experiment on DS2-V2

Cl	Acc	Pre	Rec	F1	Time (seconds)	
					Training	Testing
LR	52.6	33.4	26.4	19.8	53.707	0.070
NB	52.9	47.1	53.4	24.4	0.561	0.199
KNN	99.8	99.6	99.7	99.6	8488.109	13.189
DT	99.9	98.2	98.2	98.2	36.960	0.083
AB	12.2	61.2	35.3	20.5	296.122	10.937
RF	99.1	99.5	97.8	98.6	881.833	9.363
RSVM	100	100	100	100	864,000.140	351.150
LSVM	100	100	100	100	777,600.180	264.500